Certified reliability for any routing solver: screening buffered plans against travel-time uncertainty, with evidence from real delivery routes

Suhan Ali
Independent researcher, Lahore, Pakistan
E-mail: syedsuhanali9@gmail.com
ORCID: 0009-0004-5079-7684

Corresponding author: Suhan Ali (address as above).
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Use of generative AI. Anthropic's Claude and OpenAI's ChatGPT were used to write and debug the simulation and analysis code, to review that code, to run and check the statistical analyses, to search the literature, and to draft and revise the text of the manuscript, including its wording. The research question, the study design and the confirmatory protocols were the author's. Every experiment was run from the archived code, every reported number is regenerated from the archived records, and the author takes full responsibility for the content. **Biographical note.** Suhan Ali is an independent researcher

based in Lahore, Pakistan, working on vehicle routing and delivery planning under travel-time uncertainty. Alongside this research, Suhan is developing RouteOps, a route-planning system built on Google OR-Tools, and is studying A-Level Further Mathematics.